1. Household Robots

What Are Household Robots?

Household robots are machines designed to assist people with everyday domestic tasks. One of the most successful examples is the robotic vacuum cleaner, which automatically cleans floors inside homes. Robotic vacuum cleaners have become an important example of domestic robotics, with millions of units sold worldwide. However, developing a successful household robot is not only about making it perform its task correctly. It must also adapt to the environment and the needs of the people using it.

How Do They Work?

Robotic vacuum cleaners use navigation systems to move through indoor environments and clean floors.

Their performance depends on several important factors:

- **Navigation:** The robot must move through the environment and cover the areas that need cleaning.
- **Mapping:** Technologies that represent the surrounding environment can help improve navigation and energy efficiency.
- **Energy consumption:** Different navigation technologies and design choices influence how much energy the robot uses.
- **Cleaning performance:** The robot must clean effectively while operating within its technical limitations.

The choice of navigation technology can therefore influence both the robot's cleaning performance and its energy consumption.

Real-World Research: EPFL Study

Researchers at the École Polytechnique Fédérale de Lausanne (EPFL) investigated how robotic vacuum cleaners perform in real households. Their research combined two different studies. In the first study, researchers analyzed seven robotic vacuum cleaners to understand how technologies such as navigation systems affect technical performance and energy consumption. In the second study, they observed the use of robotic vacuum cleaners in nine households to understand users' needs and how robots fit into everyday domestic routines. By combining technical evaluations with observations of real users, the researchers investigated both the performance of the robots and their integration into people's homes.

Benefits and Challenges

Robotic vacuum cleaners can help people perform routine cleaning tasks automatically. However, the EPFL research identified several important challenges. One challenge is energy efficiency. The way a robot navigates can influence its energy consumption and cleaning performance. Another challenge is user acceptance. A robot may work correctly from a technical perspective but still fail to meet the expectations of its users. The researchers found that users' needs and perceptions were not always sufficiently considered in robot design. Robots may also be rejected when they do not fit well into people's existing domestic environments and routines.

The Future of Household Robots

The EPFL study suggests that improving domestic robots requires more than developing better technology. Researchers and designers should consider both technical performance and human needs. Improvements in navigation, mapping, and energy efficiency could make robotic vacuum cleaners more effective. At the same time, understanding how people interact with robots is essential for creating machines that integrate successfully into everyday life.

Source: https://www.sciencedirect.com/science/article/pii/S0921889013001899

2. Warehouse Robots

What Are Warehouse Robots?

Warehouse robots are automated machines used to transport, organize, and handle goods inside storage and distribution facilities. As online shopping and global logistics continue to grow, companies increasingly use robots to support warehouse operations. These machines can move products between different locations, assist workers with repetitive tasks, and improve the organization of warehouse activities.

How Do They Work?

Warehouse robots use a combination of hardware and software to operate in structured indoor environments.

- **Sensors:** Detect obstacles, workers, and other objects.
- **Navigation systems:** Help mobile robots determine their position and move through warehouse areas.
- **Control software:** Coordinates robotic tasks and movements.
- **Artificial intelligence:** Can help robots interpret their surroundings and adapt to changing situations.

Some warehouse robots follow predefined routes, while more advanced autonomous mobile robots can navigate dynamically around obstacles.

Real-World Example: Amazon Robotics

Amazon uses robotic systems in its fulfillment centers to support the movement and processing of packages. One example is Proteus, an autonomous mobile robot designed to transport heavy carts through warehouse environments. Unlike some traditional warehouse robots that operate in separated areas, Proteus is designed to navigate in spaces shared with human workers. It uses advanced sensing and navigation technology to detect obstacles and move safely through its working environment.

Benefits and Challenges

Warehouse robots can increase operational efficiency, reduce repetitive physical work, and support the movement of large quantities of goods. They can also help employees focus on tasks that require human judgment and problem-solving. However, implementing warehouse robotics requires significant investment, technical maintenance, and careful coordination between machines and human workers.

Emanuele Tonazzo

The Future

Warehouse automation is expected to become more flexible as robotic systems improve. Future warehouses may use increasingly autonomous robots that can coordinate with each other, adapt to changing layouts, and perform a wider range of logistics tasks.

Source: https://www.aboutamazon.com/stories/amazon-robotics-autonomous-robot-proteus-warehouse-packages

3. Industrial Robots

What Are Industrial Robots?

Industrial robots are programmable machines designed to perform tasks in manufacturing environments. They are commonly used in factories to assemble products, move materials, weld components, and paint surfaces. Unlike robots operating in unpredictable outdoor environments, many industrial robots work in structured spaces where tasks and movements can be precisely controlled. Their main purpose is to improve manufacturing productivity, precision, and consistency.

How Do They Work?

Industrial robots combine mechanical components, electronic systems, and specialized software.

- **Robotic arms:** Move tools or objects using multiple joints and motors.
- **Controllers:** Execute programmed instructions and coordinate movements.
- **Sensors:** Monitor positions, forces, or environmental conditions.
- **Vision systems:** Allow certain robots to identify and locate objects using cameras.

Many industrial robots perform predefined sequences of movements. More advanced systems can use sensor feedback to adjust their actions.

Real-World Example: ABB Robotics

ABB is a major manufacturer of industrial robotic systems used in automotive, electronics, and other manufacturing industries. Its robots can perform activities such as welding, painting, assembly, and material handling. For example, robotic arms can be programmed to weld components in a car factory, repeating the same operation with high precision. ABB also develops collaborative robots, known as cobots, which are designed to work alongside humans when appropriate safety requirements are met.

Benefits and Challenges

Industrial robots can perform repetitive tasks with high accuracy and consistency. They can improve production efficiency and reduce human exposure to certain hazardous working conditions. However, robotic systems can be expensive to install and maintain. They also require programming, technical expertise, and appropriate safety measures. Traditional industrial robots may need to be reprogrammed when manufacturing requirements change.

The Future

Industrial robotics is becoming more flexible through advances in artificial intelligence, sensors, and collaborative technology. Future factories may combine traditional automation with intelligent robots capable of adapting to different products and working more effectively alongside human employees.

Source: https://www.abb.com/global/en/areas/robotics/applications

4. Hospital Robots

What Are Hospital Robots?

Hospital robots are robotic systems designed to assist healthcare professionals with medical, logistical, and administrative activities. They can operate in structured indoor environments such as hospitals, clinics, and healthcare facilities. Different types of hospital robots perform different tasks. Some transport medical supplies, others assist with cleaning and disinfection, while specialized robotic systems support surgical procedures. One important application is hospital logistics, where robots can automate the transportation of materials between departments.

How Do They Work?

Autonomous hospital transport robots combine several technologies to navigate indoor environments.

- **Sensors:** Detect walls, equipment, people, and obstacles.
- **Navigation software:** Helps robots determine routes through corridors and hospital departments.
- **Digital maps:** Represent the hospital layout and support movement between locations.
- **Communication systems:** Allow robots to interact with compatible hospital infrastructure, such as automatic doors and elevators.

Some robots can automatically transport supplies to designated locations without requiring a staff member to accompany them.

Real-World Example: Aethon TUG

Aethon TUG is an autonomous mobile robot designed to transport materials inside healthcare facilities. It can be used to move medical supplies, medications, meals, and other items between hospital departments. The robot navigates through corridors using sensors and mapping technology. In suitably equipped buildings, it can also interact with elevators and automatic doors. By automating routine transportation tasks, systems such as TUG can allow hospital staff to spend more time on activities involving direct patient care.

Benefits and Challenges

Hospital robots can reduce the time employees spend transporting materials and improve the organization of internal logistics. They can also support consistent delivery routines and reduce repetitive physical work. However, hospitals are busy environments where robots must navigate

around patients, visitors, and medical staff. Reliable navigation, maintenance, hygiene, and integration with existing hospital systems are important challenges.

The Future

Hospital robots may become increasingly common as healthcare facilities adopt more automation. Future systems could offer improved navigation, greater flexibility, and better coordination with hospital infrastructure. Rather than replacing healthcare professionals, these robots could support them by handling routine logistical tasks and allowing more attention to be dedicated to patient care.

Source: https://aethon.com/